EVOLUTION SIMULATOR / ARTIFICIAL ECOSYSTEM PROJECT

Background: I am an 18-year-old student who completed Class 12 (PCMC: Physics, Chemistry, Mathematics, Computer Science) and am interested in pursuing Computer Science Engineering, particularly at Korean SKY universities (Yonsei University, Korea University, and possibly SNU).

I scored 88/100 in Computer Science and have experience with Python from the CBSE curriculum. I have also worked on a Student Record Management System project in school. My current goals are: 1. SAT preparation. 2. Mathematics improvement examination. 3. Building a substantial independent Computer Science project for learning and university applications.

Why I Chose This Project: Initially I considered several project ideas:

- Civilization Simulator
- Maritime Trade Simulator
- Human Potential Simulator
- University Decision Analyzer
- Gym Progress Simulator

After discussion, I selected an Evolution Simulator because it provides the best balance between:

- Ambition
- Scientific curiosity
- Feasibility
- Learning value
- Time available

A Civilization Simulator would likely become too large and complex for a first major independent project.

Core Philosophy: This is not intended to be just a game.

It is intended to be a computational experiment or artificial ecosystem.

The project is based on a question-driven approach:

"What happens if I change the rules of the system?"

Examples:

- Does higher speed evolve naturally?
- How does food scarcity affect evolution?
- How does mutation rate affect survival?
- Can cooperation emerge?
- Can communication emerge?
- Is there an optimal speed for survival?
- How do environmental constraints influence populations?

The goal is not to hard-code answers.

The goal is to define rules and observe outcomes.

In other words:

Build a world. Define rules. Observe consequences.

Influences: The project shares some conceptual similarities with Conway's Game of Life because simple rules can generate complex outcomes.

However, this project differs because organisms can possess traits, consume resources, reproduce, mutate, and potentially evolve more advanced behaviors.

Long-Term Vision: The simulator may eventually include:

- Food
- Energy
- Hunger
- Death
- Reproduction
- Mutation
- Environmental zones
- Communication
- Cooperation
- Resource storage
- Specialized behaviors

However, development will occur gradually, one feature at a time.

Current Development Philosophy: Add one feature. Understand it. Test it. Observe results. Only then add another feature.

The project should remain:

Scientific + Visual

rather than:

Purely Scientific or Purely Game-like

Desired final presentation:

- Moving organisms
- Food sources
- Statistics panel
- Population graphs
- Evolution tracking
- Experimental results
- Screenshots and documentation

Current Progress:

Development Environment:

✓ Python 3.11 installed ✓ Pygame installed ✓ VS Code installed ✓ Python extension installed ✓ Project folder created ✓ Project successfully runs inside VS Code

Current Project Structure:

EvolutionSimulator/ └── Evolution_simulator.py

Current Features Implemented:

✓ Organisms spawn at random positions. ✓ Organisms have:

- x position
- y position
- speed
- energy

✓ Organisms move randomly. ✓ Organisms are rendered as blue circles. ✓ Code has been cleaned and organized using:

- Constants
- Functions
- Better structure

Current Development Roadmap:

Step 1: ✓ Confirm VS Code runs project

Step 2: ✓ Clean and organize code

Step 3: Food system (Current step)

Step 4: Collision detection and eating

Step 5: Energy consumption

Step 6: Death mechanics

Step 7: Reproduction

Step 8: Mutation

Step 9: Statistics and graphs

Step 10: Experiments and analysis

Current Food Implementation Status:

Food design chosen:

Version A: Random food distribution

Food properties:

- x coordinate
- y coordinate

Food behavior:

- Static
- Does not move

Food currently exists as green dots and will later be consumed by organisms.

Expected University Application Outcome:

The project is intended to demonstrate:

- Programming ability
- Independent learning
- Scientific thinking
- Systems modelling
- Computational experimentation
- Curiosity-driven research

The project's value comes not only from coding, but from designing experiments, testing hypotheses, analyzing results, and explaining conclusions.

Personal Motivation:

One of my strongest interests is asking "what if?" questions about systems.

Examples include:

- Physical systems
- Evolutionary systems
- Complex systems
- Civilization-level systems

This project allows me to turn those questions into something measurable and observable through code.

The ultimate goal is to build a world, define its rules, and discover what emerges from them.